ALICE L WALTON SCHOOL OF MEDICINE

Aortic Regurgitation and Pulmonary Valve Disease

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Adapted from: Aortic Regurgitation

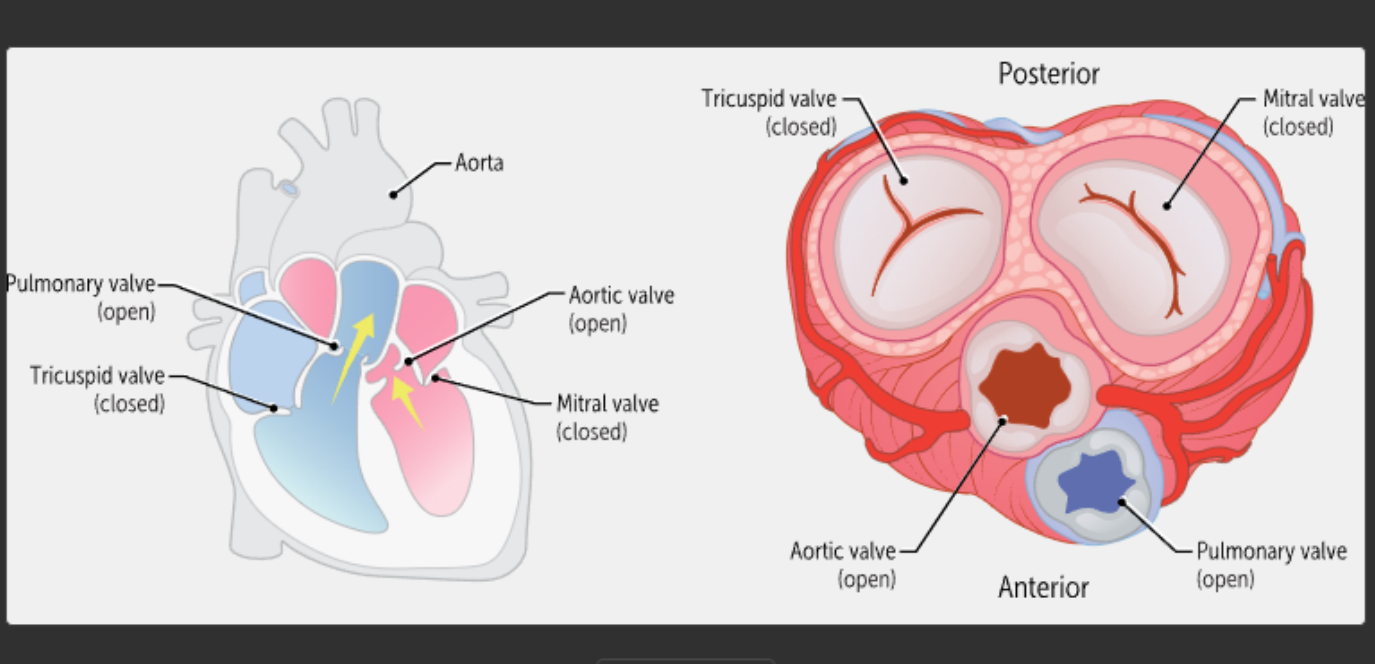
Listen to this Brick
0:00 / 26:55 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe the etiologies and pathophysiology of acute and chronic AR
2. Describe the clinical presentation, diagnosis, and management of AR.
3. Discuss Pulm… …ologies and

aorta. During this event the tricuspid and mitral valves remain closed (Figure 1).



QUIZ

Figure 1 Blood flow from left ventricle to aorta (left) and cross section showing cardiac skeleton (right)

When the LV relaxes in diastole, the three leaflets of the aortic valve slam together and stop the pressurized blood in the aorta from shooting back into the heart. AR happens when the three aortic leaflets can't close all the way, letting blood flow back from the aorta into the heart during diastole (Figure 2).

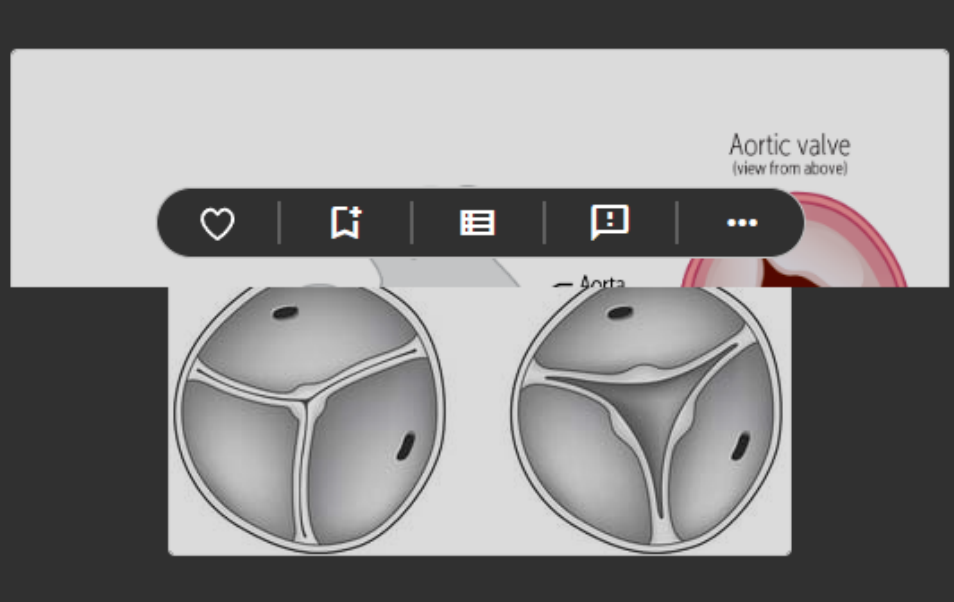


Figure 4. Dilation of the aortic root will ultimately prevent valve cusps from coapting.

Etiologies for AR from aortic valve disease include disorders that more commonly cause Aortic Stenosis but can also cause AR:

- Calcific valve disease from old age
- Bicuspid aortic valve
- Rheumatic heart disease

What is the most common cause of aortic regurgitation (AR) in the United States?

Figure 5. Pressure–volume loop of aortic regurgitation (left). Aortic and LV pressure curve (right). Note the rapid decrease in aortic diastolic pressure which will result in a decrescendo murmur.

The increased stroke volume allows most AR patients to remain asymptomatic for many years. In contrast, patients with more severe chronic AR become symptomatic when the effects of chronic volume overload exceed the limits of compensation of the Frank-Starling mechanism. Further volume overload results in a backup of blood volume and pressure potentially resulting in heart failure (Figure 6).

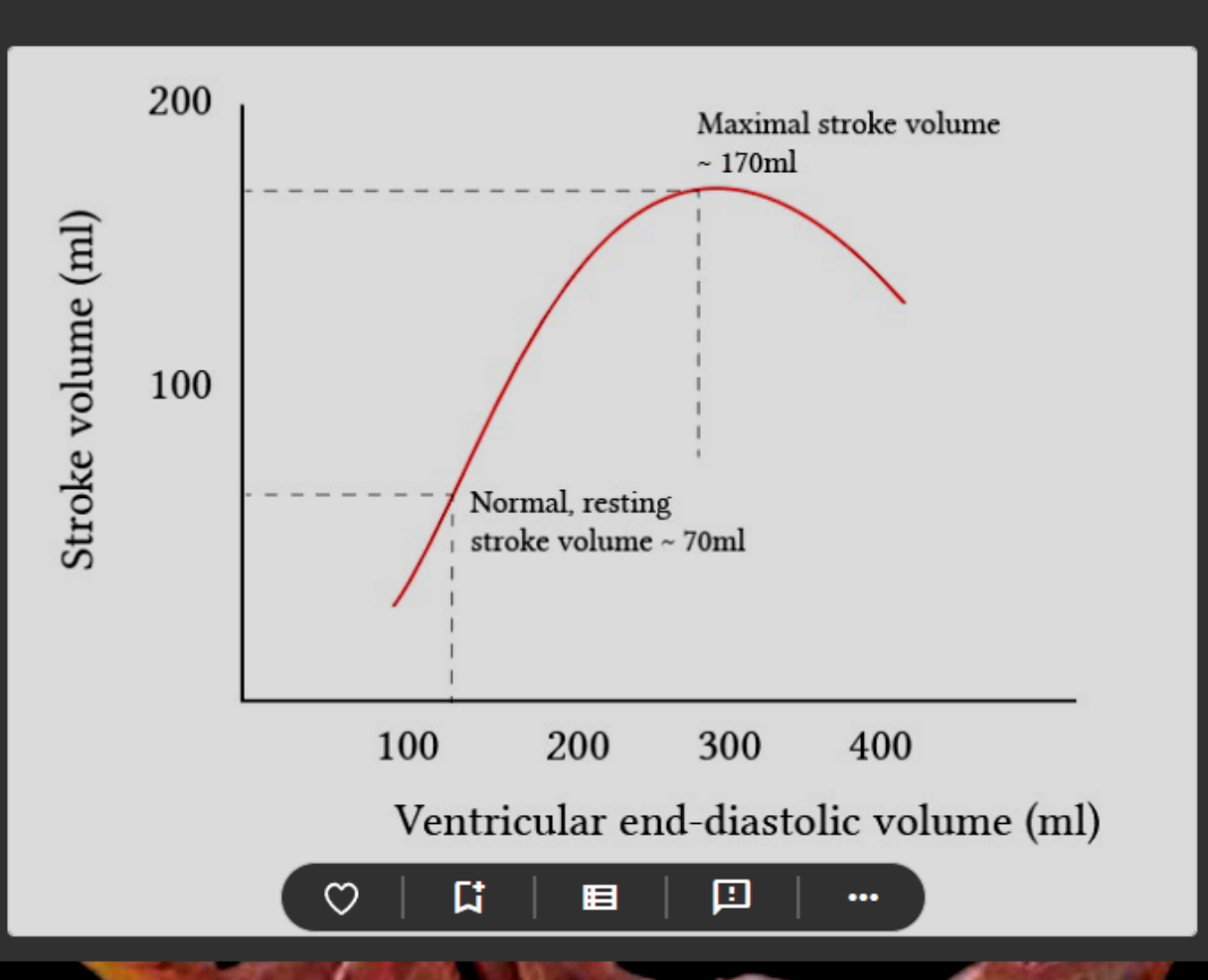
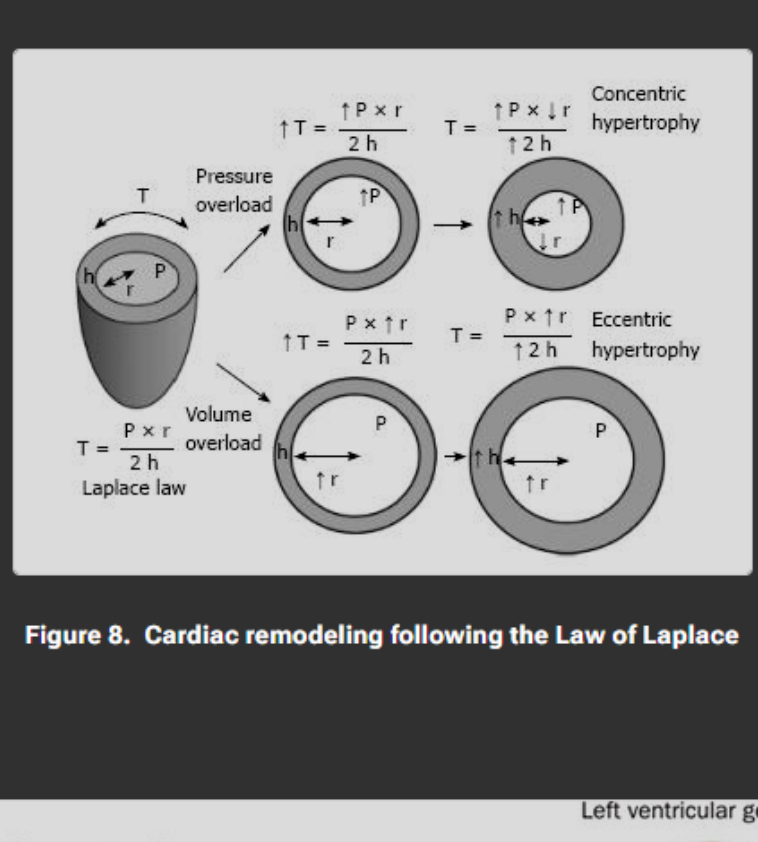


Figure 7. Left ventricular eccentric hypertrophy in dilated cardiomyopathy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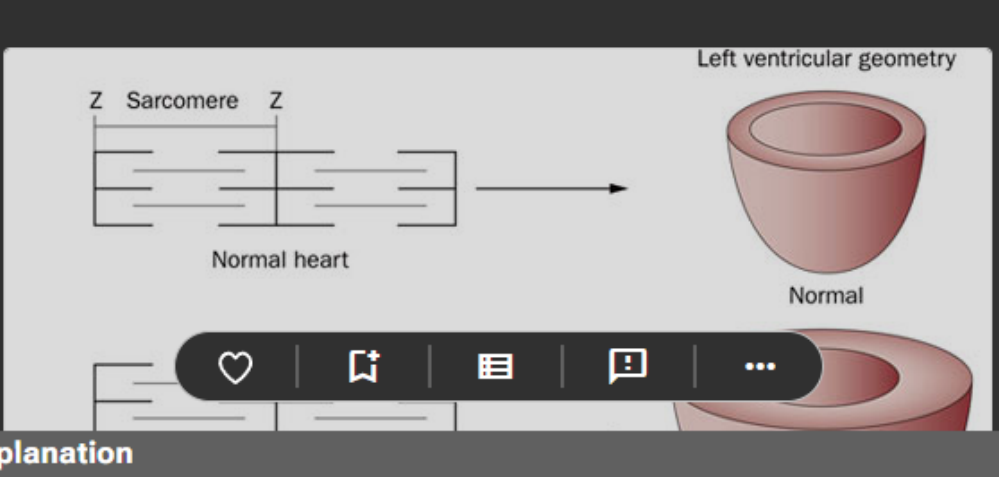
$T = \frac{P \times r}{2h}$
Laplace law
Pressure overload
Volume overload
$\uparrow T = \frac{\uparrow P \times r}{2h}$
$T = \frac{\uparrow P \times \downarrow r}{\uparrow 2h}$
Concentric hypertrophy
$\uparrow T = \frac{P \times \uparrow r}{2h}$
$T = \frac{P \times \uparrow r}{\uparrow 2h}$
Eccentric hypertrophy

Figure 8. Cardiac remodeling following the Law of Laplace



Explanation

The correct answers are (C, D). AR is a volume overload, leading to eccentric hypertrophy. This involves an increasing LV volume with less of an increase in wall thickness and ultimately a decreased ejection fraction (HFrEF). AS is a pressure overload, like hypertension. This involves an increase in LV wall thickness and ultimately impaired relaxation (diastolic dysfunction) and preserved contractility (HFpEF).

Left Ventricular heart failure can cause dyspnea and pulmonary edema as the pressure increases in the LV during end-diastole, increasing pulmonary venous pressure. Also, microvascular changes can result in decreased perfusion and myocardial ischemia.

How Do Patients With Aortic Regurgitation Present? (LO 2)

Many patients with chronic AR will present with no symptoms. Patients who have either acute AR or more severe chronic AR with heart failure can have a mixture of symptoms that may be attributed to many other causes:

- Shortness of breath (dyspnea), especially with lying down (orthopnea) or sleeping (paroxysmal nocturnal dyspnea)
- Palpitations
- Exercise intolerance
- Chest pressure o… …ent blood pressure

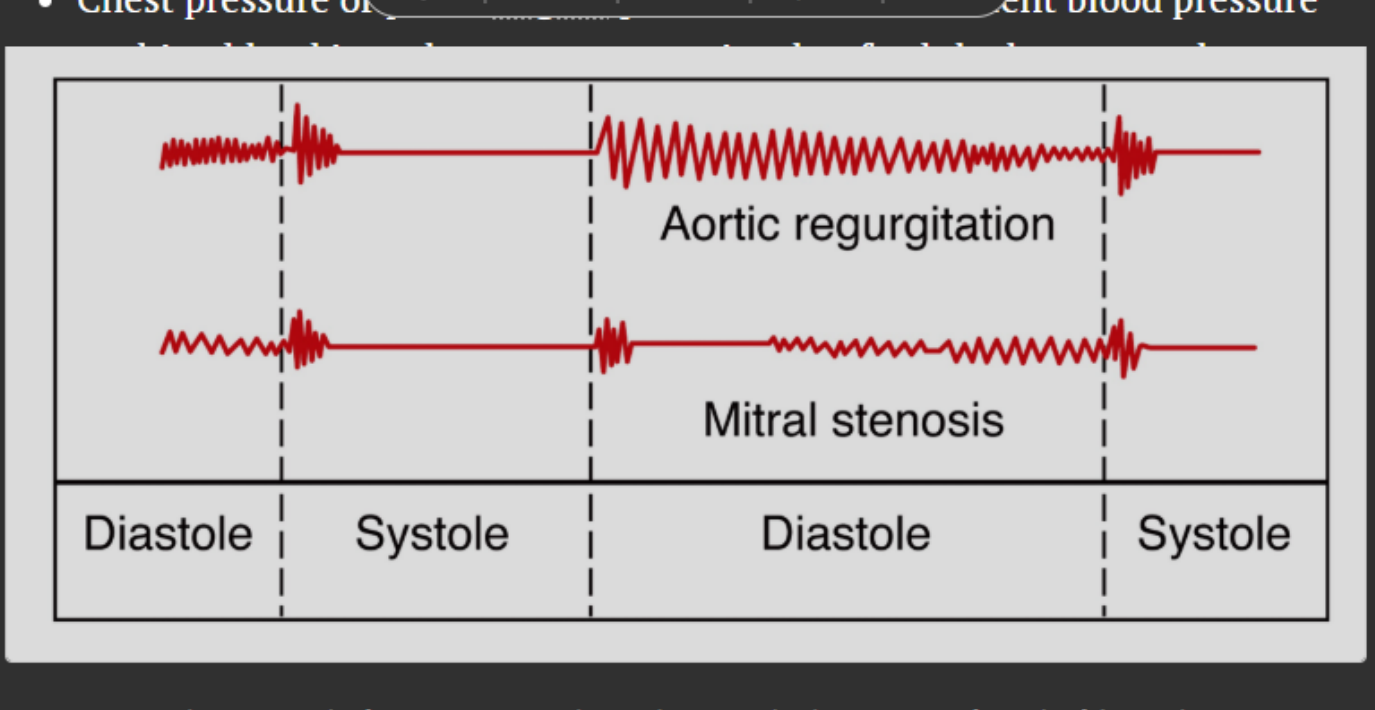


Figure 7. Typical murmur pattern in Aortic Regurgitation compared to Mitral Stenosis

The Aortic Regurgitation murmur is best auscultated at the parasternal border of either the right 2nd intercostal space or left 3rd intercostal space (the Erb point) at the end of expiration with the patient sitting up and leaning forward (Figure 8). Expiration decreases lung volume and increases venous return to the left heart (but decreases venous return to the right heart). The murmur is characterized as a soft, high-pitched, early diastolic, decrescendo (decreasing in intensity as the diastole progresses) murmur. It is decrescendo because the regurgitant flow rate is highest when aortic diastolic pressure is highest and decreases with the rapidly falling aortic diastolic pressure as regurgitation continues.

An Austin-Flint murmur may also be heard, which is a low, rumbling diastolic murmur best heard at the apex due to the regurgitant vent vibrating against the mitral valve (the valve which is best heard at the apex).

Inspiration increases right sided murmurs due to increased venous return to the right heart from low intrathoracic pressure, but decreases left sided murmurs due to increased blood pooling in the expanded lungs.

Expiration increases left sided murmurs due to decreasing lung volume which increases left sided venous return, but lessens right sided murmurs due the higher intrathoracic pressure decreasing right sided venous return.

Differentiation of common heart murmurs from Hypertrophic Cardiomyopathy (HCM) and Mitral Valve Prolapse with Mitral Regurgitation (MVP with MR):

Standing upright and the Valsalva maneuver both decrease Venous Return to right and left heart: both decrease most heart murmurs. Exceptions include disorders worsened by lower LV volume (diameter). The murmur of Hypertrophic Cardiomyopathy (HCM) increases at lower LV volumes when the LV walls are closer together and more obstructive. Mitral valve Prolapse (MVP) with Mitral Regurgitation (MR) is also an exception, as the lower volumes change papillary muscle position and loosen the chordae, making prolapse occur earlier and MR occur louder and longer (begins earlier with the prolapse).

Squatting and passive leg raise while supine both increase Venous Return to right and left heart: both increase most heart murmurs. Again the exception is HCM because a larger LV volume decreases the obstruction. MVP with MR is also an exception, as the higher volumes tighten the chordae and make prolapse occur later and MR occur quieter and more brief (begins later with the prolapse).

Differentiation of Left heart Stenotic vs Regurgitant Murmurs

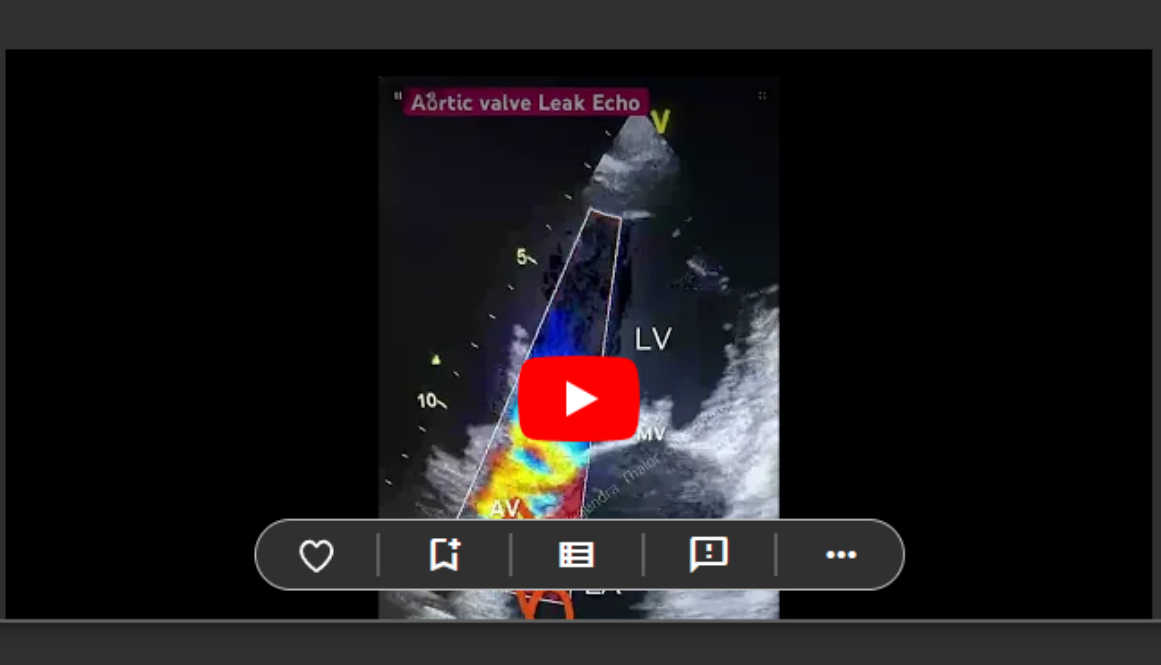
Decreased afterload allows AS to eject better, increasing flow and murmur volume. Allows HCM, MVP/MR to eject better, decreasing volume and increasing murmur volume. Allowing better outflow, it will decrease Left sided regurgitant murmurs.

Decreased Afterload: Inhaling Amyl Nitrate, a potent vasodilator. Increases AS

Imaging

AR is best diagnosed with **echocardiography.** This may also reveal the etiology of the AR such as thickening of the aortic valve leaflets, dilated aortic root, vegetations, calcification, bicuspid aortic valve, or flail aortic leaflets (leaflets that flip back into the LV during diastole). If the AR is severe enough, the echocardiogram may also show a dilated LV with abnormally large end diastolic and end systolic volumes.

Doppler echocardiography is also used to determine the amount of flow through the aortic valve back into the LV (up towards the probe), as shown below in the apical view echo below. The extent of the regurgitation into the LV can be quantified. The below video has no sound.



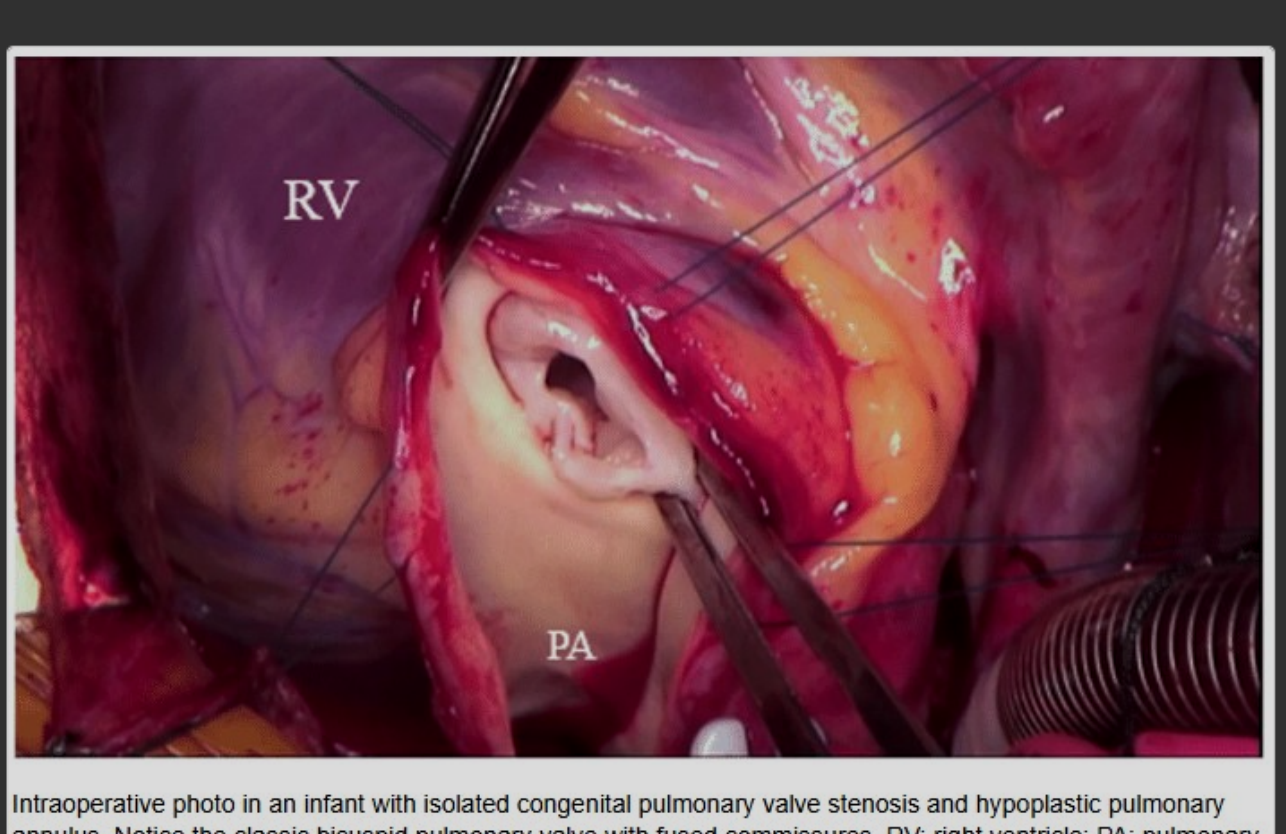
Medical Therapy

For patients with more advanced symptoms, treatment begins with decreasing the systemic blood pressure, which will reduce the afterload and therefore the amount of blood that regurgitates back into the LV during diastole. **Afterload-reducing agents** include angiotensin-converting enzyme (ACE) inhibitors, angiotensin-receptor blockers (ARBs), and dihydropyridine calcium channel blockers like nifedipine.

Direct vasodilators like hydralazine can be used in chronic AR, but not in acute AR. Because their rapid action does not allow adaptation, vasodilation will drop blood pressure triggering an increase in heart rate which shortens diastole and impairs ventricular filling, potentially worsening the regurgitation.

How do ACE inhibitors help patients with chronic AR?

etiologies of pulmonary valve disease, the pressure differential is too low on the right side of the heart, resulting in quieter murmurs.



Intraoperative photo in an infant with isolated congenital pulmonary valve stenosis and hypoplastic pulmonary annulus. Notice the classic bicuspid pulmonary valve with fused commissures. RV: right ventricle; PA: pulmonary artery.

Figure 9. Pathologic, congenital bicuspid Pulmonary Valve with fused commissures causing Pulmonary Stenosis

Because the right heart and pulmonary system is low pressure, mild/moderate cases of PS and PR are well tolerated. In cases of severe valve disorder (often with congenital PS), right sided heart failure can result. Classic signs and symptoms of right heart failure include jugular venous distention, peripheral edema, liver swelling,

leaking valve, which can lead to eccentric hypertrophy with dilation of the ventricle.
- Lack of coronary flow during diastole in patients with AR can lead to myocardial ischemia.
- AR causes an early diastolic decrescendo murmur due to the rapidly falling aortic diastolic pressure, and is heard with a stethoscope at the parasternal left second or right third intercostal space on the chest.
- Wide pulse pressure leads to a "water-hammer pulse" (Corrigan pulse) on physical exam.
- AR is diagnosed with an echocardiogram, and Doppler echocardiography is used to determine the amount of regurgitation through the aortic valve.
- For asymptomatic patients, the management is watchful waiting with repeat echocardiograms, the timing of which depends on AR severity.
- In symptomatic AR, management centers on decreasing blood pressure with medications like angiotensin-converting enzyme (ACE) inhibitors.
- The only definitive treatment for patients with symptoms of AR is aortic valve replacement.
- The pulmonary valve is least likely to harbor disease due to the low pressure in the pulmonary circulation and when right sided valve disease occurs (carcinoid syndrome, infective endocarditis), it affects the Tricuspid Valve much more commonly.
- The most common pulmonary valve disorder is congenital Pulmonary Stenosis.

Review Questions

1. A 65-year-old pati… …s presents to clinic with slowly worsening shortness of breath in the middle of the night and aortic regurgitation confirmed on echocardiography with preserved ejection fraction. Which of the following is an associated physical exam finding and recommended management?

A. ○ crescendo-decrescendo murmur, aortic valve replacement
B. ● forceful, rapidly diminishing carotid pulse, observation with office and echocardiography follow up
C. ○ increased murmur with inspiration, best treated with diuretics
D. ○ radiation of the murmur to the carotid arteries, balloon dilation

Hide Explanation

The correct answer is (B), a water hammer pulse. In the setting of minimal symptoms it is best managed by office and echo follow up. The AR murmur is decrescendo; Aortic Stenosis causes a crescendo-decrescendo murmur. Aortic valve replacement is reserved for severe symptoms or LV dysfunction (A). The AR murmur decreases with inspiration, and diuretics are not typically used in AR unless the patient has volume overload (C), (but they are used to treat Tricuspid Regurgitation and Mitral Valve disease). Afterload reducing agents like ACE Inhibitors are the initial treatment for more severe AR symptoms because they lessen the regurgitant volume. The murmur of AS radiates to the carotids, and valvular stenosis may be treated by balloon dilation (D).

3. In which of the foll… …ikely to find isolated pulmonary valve disease?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the etiologies and pathophysiology of acute and chronic AR~~
02 ☑ ~~Describe the clinical presentation, diagnosis, and management of AR.~~